Response to Technical Inspection Comments
Manuscript ID: [TBD]
Adaptive Multi-Objective Chaotic S-Box Design for Lightweight and Secure IoT Communication

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Dear Editor,

Thank you for the thorough technical inspection of our manuscript. We have carefully addressed each comment raised during the inspection. Below, we provide a point-by-point response detailing the changes made to the manuscript. All changes are marked in the revised manuscript file `main_Scientific_Reports.tex` and reflected in the compiled PDF `main_Scientific_Reports.pdf`.

Summary of Revisions

Comment #	**Description**
1	Please provide a Funding Declaration in the manuscript. If there was no Funding, please state so.
2	We request authors to provide the same Data Availability statement from the manuscript and update in the "Submission system".

Table 1: Technical inspection comments addressed in this revision.

Point-by-Point Response

Comment 1: Funding Declaration

Editor Comment

Please provide a Funding Declaration in the manuscript. If there was no Funding, please state so.

Our Response

We confirm that this research did not receive any external funding from any funding agency, grant, or individual. No financial support was obtained from governmental, institutional, or private sources for the conduct of this study, the preparation of the manuscript, or the decision to submit it for publication.
To address this comment, we have added a dedicated **Funding** section to the manuscript, placed between the *Acknowledgements* and *Author information* sections. The section contains the standard Nature Portfolio declaration for unfunded research:

> **Funding**
> This research received no external funding.

Location in the revised manuscript: The Funding declaration appears in the unnumbered *Funding* section after Acknowledgements and before Author information in `main_Scientific_Reports.tex` and the compiled manuscript PDF.
Verification: The declaration reads: "This research received no external funding."

Comment 2: Data Availability Statement in Submission System

Editor Comment

We request authors to provide the same Data Availability statement from the manuscript and update in the "Submission system".

Our Response

We have revised the manuscript Data Availability section to identify the public GitHub repository and to state precisely that Zenodo DOI 10.5281/zenodo.23439389 refers to the earlier v1.0.0 snapshot, not the later source and paired results now reported. The final revised materials are available on the repository's current `main` branch. Before resubmission, the corresponding author must update the submission-system Data Availability field to "YES" and paste the exact current manuscript statement; this external form update cannot be made from the manuscript files.

Revision status

The latest manuscript PDF is 11 pages and contains 9 cited references, 7 figures, and 1 table (8 display items). The Funding declaration is present. The Data Availability section cites the public GitHub repository and identifies Zenodo v1.0.0 as an earlier snapshot; an updated Zenodo version/DOI and submission-system field synchronization remain necessary before resubmission. The response should be finalized after those external steps are completed.

Sincerely,

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(on behalf of all co-authors)
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